

Hackathon Problem Statements

Regd No: 25091A32M6, 25091A32N6

Problem Statement No: 1

Crop Yield Prediction System Design a system that predicts crop yield for a given region using historical rainfall, temperature, soil pH, and fertilizer usage data. The system should apply regression models (linear / polynomial) from the data science pipeline to estimate output per acre, while an underlying Java OOP module manages farmer profiles, land records, and crop history using class hierarchies. The tool should help agricultural officers plan procurement and storage in advance, reducing post-harvest wastage. Visualize predicted vs actual yield trends over multiple seasons to build farmer trust in the system.

Regd No: 25091A32Q0, 25091A3251

Problem Statement No: 35

Health Camp Resource Allocator Design a resource allocation tool for free health camps that uses knapsack-style optimization to allocate a limited budget across medicines, staff, and equipment based on expected patient demographics and disease prevalence in the target area. Use OOP to model resource categories with cost-benefit attributes. This helps NGOs and government bodies maximize health impact per rupee spent in underserved rural and tribal areas.

Regd No: 25091A3201, 25091A3232

Problem Statement No: 41

Urban Flood Risk Predictor Build a flood risk prediction model using regression analysis on rainfall intensity, drainage capacity, and elevation datasets to identify flood-prone urban zones before monsoon season. Visualize risk levels on a city map to guide preventive drainage maintenance and public advisories. Use OOP to structure zones, drainage infrastructure, and risk assessments. This is critical for cities facing recurring urban flooding due to poor drainage planning and unplanned construction.

Regd No: 25091A32D8, 25091A3280

Problem Statement No: 81

Coastal Erosion Prediction System Build a coastal erosion trend predictor using regression analysis on historical shoreline position datasets to project future erosion rates for vulnerable coastal stretches. Visualize risk zones to support coastal zone management planning. Use OOP to structure coastal segments, historical measurements, and risk projections. This is highly relevant for India's extensive and increasingly vulnerable coastline communities.

Regd No: 25091A32C1, 25091A32B9

Problem Statement No: 86

Forest Fire Spread Predictor Create a forest fire spread simulation using a cellular automaton model combined with graph-based propagation logic, factoring in wind direction, vegetation density, and terrain slope from public datasets to predict fire spread patterns. Use OOP to model terrain cells and fire-state transitions. This supports forest departments in fire response planning and resource pre-positioning during high-risk seasons.

Regd No: 25091A32H3, 25091A32G2

Problem Statement No: 98

Insurance Premium Calculator Create an insurance premium calculation engine using an actuarial rule-based decision system (OOP design) that computes premiums based on risk factors like age, health indicators, and coverage type, similar to real insurance underwriting logic. This helps demystify insurance pricing for consumers and can serve as a foundation for comparing policy options transparently.